AgileFlow AI

Human-in-the-Loop Multi-Agent Framework for Agile SDLC Automation

TAF / Project Scope Summary - Knowledge Passing Document

1. Project Identity

Project ID	J26-SE-358
Official Project Title	AgileFlow AI - Human-in-the-Loop Multi-Agent Framework for Agile SDLC Automation
Research Area	Software Systems and Technologies / Software Engineering
Primary Research Focus	Human-in-the-loop multi-agent automation of Agile SDLC activities with traceability and validation evidence
Prototype Type	Web application prototype with orchestrated agent workflow, approval gates, shared knowledge base, and final reports
Main Scope Boundary	The system creates a structured Agile project environment and validated runnable prototype foundation; it is not claimed as a complete production-ready software builder.

2. Purpose of This Document

This document is the first knowledge-passing file for new ChatGPT chats, AI coding agents, UI designers, and implementation team members. It summarizes the approved research scope, architecture, component responsibilities, expected outputs, human approval points, and implementation boundaries of AgileFlow AI.

Use this document as the base context before generating UI designs, API plans, folder structures, AI prompts, implementation tasks, or presentation content.

3. Problem Statement

Small software teams, start-ups, freelance teams, and student project groups often struggle to convert raw client ideas into a connected Agile SDLC workflow. Client inputs may come through text, documents, audio, video, web links, or interviews, but they are often unclear, incomplete, duplicated, or conflicting.

Existing tools support individual tasks such as project management, design, coding, testing, and security checking, but they usually do not connect the complete SDLC. This causes weak traceability, disconnected artifacts, late testing, missing security validation, and reduced human control over AI-generated outputs.

4. Research Gap

- Requirement tools rarely connect directly with design, coding, testing, security validation, and final evidence reports.
- Design and planning tools often generate artifacts without strong requirement-to-design-to-test traceability.
- AI coding tools usually focus on code generation, not full SDLC workflow control, approvals, Git automation, execution logs, and validation loops.
- Testing and security tools are often used after implementation, creating late defect discovery and weak correction tracking.
- Current solutions do not sufficiently combine multi-agent automation, human approval gates, orchestrator-controlled feedback loops, and a shared knowledge base in one Agile SDLC workflow.

5. Proposed Solution

AgileFlow AI proposes a human-in-the-loop multi-agent web application framework that connects the Agile SDLC from raw client input to validated prototype foundation. The system uses specialized agents coordinated by a Central SDLC Orchestrator and supported by a Shared Knowledge Base.

The system does not remove humans from the SDLC. Instead, humans review and approve important artifacts such as requirements, design/planning outputs, pre-code test specifications, generated prototype code, and final validation/security reports.

6. Core System Workflow

1. Client input is collected as text, documents, web links, audio, video, or interview responses.
2. Requirement Gathering and Analysis Agent structures the client idea into requirements, user stories, acceptance criteria, and requirement IDs.
3. Human Approval Gate H1 validates requirement outputs before design begins.
4. Project Management and Design Agent creates Agile planning and design artifacts.
5. Human Approval Gate H2 validates planning and design outputs.
6. Pre-Code Test Planning Agent creates test specifications, test skeletons, security test cases, and requirement-to-test mapping.
7. Human Approval Gate H3 validates the pre-code test package.
8. Implementation and Git Automation Agent generates selected prototype modules, manages Git metadata, and performs sandbox execution or simulation.
9. Human Approval Gate H4 validates the generated implementation.
10. Testing, Validation and Security Agent executes validation, reports bugs, performs security checking, and creates final evidence.
11. Human Approval Gate H5 validates the final results.
12. Final outputs include a structured Agile project environment, validated prototype foundation, test/security reports, bug reports, and traceability matrix.

7. Component Responsibilities

Component	Owner / Member	Main Responsibility	Key Outputs
Requirement Gathering and Analysis Agent	N A Illangasinghe - IT23391536	Collects and validates client requirements from multi-source inputs.	SRS, FR/NFR, user stories, acceptance criteria, personas, REQ IDs, quality report
Project Management and Design Agent	Kulunu Kasthuri Arachchi - IT23375628	Converts approved requirements into Agile planning and design artifacts.	Jira-ready backlog, sprint plan, UML, API spec, DB schema, UI flow, wireframes, traceability map
Implementation and Git Automation Agent	W M D D Karunarathna - IT23145320	Generates selected prototype modules and supports Git/workspace execution flow.	Project structure, generated code modules, Git metadata, build/run logs, code traceability
Testing, Validation and Security Agent	R A K Hansika - IT23140998	Generates pre-code test artifacts and validates the implementation after coding.	Test package, test results, bug reports, security findings, validation report, requirement-to-test links
Central SDLC Orchestrator	Shared / Group-level component	Controls workflow, handovers, approval gates, retry loops, routing, and state.	Workflow state, failure routing decisions, approval status, event history
Shared Knowledge Base	Shared / Group-level component	Stores approved artifacts and traceability information for all agents.	Requirements, designs, Jira items, code metadata, tests, logs, bug/security reports, traceability links

8. Human-in-the-Loop Approval Gates

Gate	Approval Point	Purpose
H1	Requirement Approval	Confirms that requirements are clear, complete, and suitable for design.
H2	Design and Planning Approval	Confirms that backlog, design, API, database, and UI artifacts are acceptable.
H3	Pre-Code Test Approval	Confirms that test specifications and expected behavior are correct before implementation.
H4	Implementation Approval	Confirms that generated prototype modules and Git/execution outputs are

		acceptable.
H5	Final Validation Approval	Confirms that test results, security results, and final evidence are acceptable.

9. Scope Boundaries and Assumptions

- The system is a research prototype, not a complete replacement for a software development team.
- The prototype should demonstrate a controlled end-to-end Agile SDLC automation workflow using selected sample domains.
- Generated implementation should be treated as a prototype foundation that developers can continue improving.
- The first implementation can use mock integrations for Jira, GitHub, Docker, testing, and security scanning before replacing them with real integrations.
- All major agent outputs must be reviewed and approved by humans before the workflow moves forward.
- The system must preserve traceability between requirements, designs, backlog items, code modules, tests, bugs, security findings, and final reports.

10. Input and Output Summary

System Inputs	System Outputs
Client idea, text descriptions, documents, web links, audio/video recordings, interview responses, human feedback, approved artifacts from previous stages.	SRS, requirements, user stories, acceptance criteria, personas, backlog, sprint plan, UML/API/DB/UI artifacts, test package, code modules, logs, reports, security findings, traceability matrix.

11. Suggested Implementation Stack

Layer	Recommended Technology	Reason
Frontend	React + Vite + Tailwind CSS	Fast dashboard development, reusable components, suitable for Figma-to-implementation flow.
Backend	Python FastAPI	Good for AI workflows, JSON APIs, modular services, and integration with LLM/tooling.
Database	MongoDB	Flexible JSON storage for requirements, designs, logs, test cases, and traceability links.
Storage	Local project storage first	Simple for prototype files, generated artifacts, logs, and reports.
AI/LLM Layer	Prompt engineering + structured JSON outputs + optional RAG	Feasible for final-year prototype without training a large model from scratch.
Validation Tools	Pytest, Playwright/Postman, Semgrep/Bandit/npm audit/Gitleaks later	Supports testing and security validation evidence.

12. Evaluation Metrics

- Requirement quality: clarity, completeness, duplicate/conflict detection, human approval score.
- Traceability coverage: percentage of requirements linked to design, backlog, code, tests, bugs, and reports.
- Design/planning validity: correctness of backlog, API, database, UI flow, and design-contract outputs.
- Implementation success: generated module completeness, build/run success, log quality, retry/self-fix success for limited errors.
- Testing effectiveness: test coverage, pass/fail rate, bug detection, regression success, evidence quality.
- Security validation: number/severity of findings, false-positive review, vulnerability fixing/re-check success.
- Human-in-the-loop effectiveness: number of approvals/rejections, correction cycles, reviewer satisfaction.

13. First Implementation MVP

The first MVP should focus on a working end-to-end workflow rather than perfect AI generation. A sample domain such as Book Management, Appointment Booking, or Tuition Management can be used for demonstration.

13. Create a project from a client idea.
14. Generate requirement artifacts and approve H1.
15. Generate design/planning outputs and approve H2.
16. Generate pre-code test package and approve H3.

17. Generate selected prototype modules with Git/log metadata and approve H4.
18. Generate validation/security reports and approve H5.
19. Display final traceability matrix and evidence reports.

14. Knowledge Package Usage Rule

When starting a new ChatGPT chat or AI coding-agent session, upload this document first. Then upload the UI/Figma design plan, architecture diagram, implementation working plan, API/data schema plan, and task assignment documents. The AI assistant should preserve this architecture and should not rename agents, change responsibilities, remove human approval gates, or replace the orchestrator-controlled workflow.

15. Quick Scope Reminder

- Official name: AgileFlow AI - Human-in-the-Loop Multi-Agent Framework for Agile SDLC Automation.
- Core idea: connect Agile SDLC stages using four specialized agents, an orchestrator, shared knowledge base, human approval gates, and traceability.
- Main output: structured Agile project environment and validated prototype foundation.
- Do not present it as a fully automatic production software builder.
- Keep all outputs traceable and human-approved.